NOETHERIAN CHOW LEMMA FOR SPACES: EMBED AFTER AN ADMISSIBLE BLOW-UP INTO A PROPER REPRESENTABLE OVER PROJECTIVE SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9224-9244 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 9246-9254, the complete author proof (7 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Noetherian avatar of the collected Nagata-compactification H3 card L16082 (chow-finite-type); the proper-representable factorization diagram is the distinct content. Borderline note: the proof is a two-step named reduction, accepted per the lane's written-reduction rule.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Limits of Spaces, lemma embedding-into-affine-over-qs (generic affine-space immersion for spaces over Noetherian bases); the chapter's lemma blowup-to-find-embedding (the blow-up improvement turning rational embeddings into diagram factorizations).

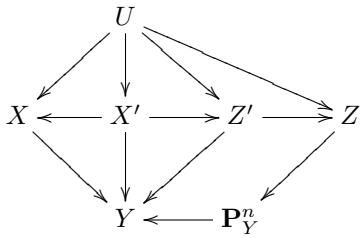
Source reference. Original lemma 088R: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9171–9179. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 088N: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9254–9254. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f separated, of finite type, and Y Noetherian. Then there exists a*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

dense open subspace $U \subset X$ and a commutative diagram



where the arrows with source U are open immersions, $X' \to X$ is a U-admissible blowup, $X' \to Z'$ is an open immersion, $Z' \to Y$ is a proper and representable morphism of algebraic spaces. More precisely, $Z' \to Z$ is a U-admissible blowup and $Z \to \mathbf{P}^n_Y$ is a closed immersion.

Proof. By Limits of Spaces, Lemma 088N there exists a dense open subspace $U \subset X$ and an immersion $U \to \mathbf{A}^n_Y$ over Y. Composing with the open immersion $\mathbf{A}^n_Y \to \mathbf{P}^n_Y$ we obtain a situation as in Lemma 088R and the result follows. □